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Consensus Guidance for Beta-Lactam Antibiotic Dose Individualization in Acutely Ill Patients

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SPECIAL RESEARCH REPORT **OPEN ACCESS**

Consensus Guidance for Beta-Lactam Antibiotic Dose Individualization in Acutely Ill Patients: Endorsed by the American College of Clinical Pharmacy, European Society of Clinical Microbiology and Infectious Diseases, European Society of Clinical Microbiology and Infectious Diseases PK/PD of Anti-Infectives Study Group, Infectious Diseases Society of America, International Association of Therapeutic Drug Monitoring and Clinical Toxicology, Society of Critical Care Medicine, and the Society of Infectious Diseases Pharmacists

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ABSTRACT

Beta-lactam antibiotics (beta-lactams) are first-line treatments for most major infectious syndromes in acutely ill patients, with in vitro activity against common pathogens, demonstrated efficacy in trials, and a perceived low risk of adverse effects. The current dosing approach, informed by population-level data, tends to be simplistically reduced to a “one size fits all” dosing method which only accounts for body weight (i.e., in pediatrics) and end organ function to estimate drug clearance (e.g., estimated glomerular filtration rate); this approach results in substantial variability in observed serum concentrations in acutely ill patients, which can compromise real-world effectiveness and safety. Beta-lactam dose individualization, therefore, defined as a dosing regimen for one patient informed by measured concentrations from that patient, has been recommended in international clinical guidelines. Comprehensive guidance on best practices for beta-lactam dose individualization is lacking. To address this knowledge gap and develop guidance, a multidisciplinary panel of international experts was assembled from the disciplines of infectious diseases, critical care, pharmacometrics, and laboratory medicine, representing both adults and pediatrics. The panel systematically evaluated the literature, using the GRADE approach where feasible, to address broad thematic questions pertaining to whether beta-lactam dose individualization should be pursued, for what indications beta-lactam dose individualization is warranted, and the most critical considerations and best practices for implementation of beta-lactam individualization. The resultant consensus recommendations will equip healthcare professionals caring for acutely ill patients treated with beta-lactam therapy with the evidence and tools necessary to guide optimal use of beta-lactam dose individualization.

Beta-lactam antibiotics (beta-lactams) are first-line treatments for patients with infections and exhibit considerable pharmacokinetic/pharmacodynamic (PK/PD) variability in acutely ill patients which results in difficult to predict effectiveness and safety. Beta-lactam dose individualization, defined in this document as a dosing regimen for one patient informed by measured concentrations from that patient, may improve PK/PD target attainment and, in turn, clinical outcomes. Beta-lactam dose individualization is recommended in contemporary international clinical guidelines (e.g., for sepsis, pneumonia) to improve antibiotic effectiveness and safety. Comprehensive guidance on the key clinical, laboratory, and practical considerations for beta-lactam dose individualization is needed.

An international multidisciplinary panel of 19 experts was convened to provide consensus recommendations on the use of beta-lactam dose individualization in acutely ill patients. These recommendations are intended for healthcare professionals caring for hospitalized patients treated with beta-lactam therapy and those aiming to implement or optimize a beta-lactam dose individualization program. The most clinically important questions were identified and prioritized by the panel and grouped into three broad thematic questions: 1) What is the clinical impact of beta-lactam dose individualization? 2) What are the minimum requirements for successful beta-lactam dose individualization? 3) What are the technical and methodological considerations necessary to achieve beta-lactam dose individualization?

A systematic review of the literature was performed by a research librarian to identify all studies relevant to the key questions. When sufficient data were available, recommendations were informed by meta-analyses and the Grading of Recommendations Assessment, Development and Evaluation (GRADE) method. An evidence-to-decision framework was applied to account for

importance and frequency of the problem, values and preferences of those affected, certainty of the evidence, health benefits and harms, burden and balance, resource implications, equity, acceptability, and feasibility. In cases where the evidence was insufficient to qualify for GRADE assessment, best practice statements were developed based on consensus of the panelists. Consensus recommendations and best-practice statements were discussed and voted on by the panel and endorsed by international multidisciplinary professional organizations.

A summary of the questions, recommendations, and best practice statements is provided in the Table 1. A detailed characterization of the methods, literature evaluation, evidence summaries and rationales for each recommendation, and proposed future directions is available in the full text of this guidance.

1 | Introduction

Beta-lactams are a first-line antibiotic treatment for most patients with serious infections [1–6]. Up to 70% of patients treated with antibiotics for sepsis receive beta-lactams [7]. Most beta-lactam dosing relies on a “one size fits all” strategy where doses of drugs are scaled to body size (e.g., weight) or estimates of drug clearance (e.g., estimated glomerular filtration rate). Generally, these dosing regimens are based on in vitro and in vivo pharmacokinetic/pharmacodynamic (PK/PD) studies, clinical PK from phase 1 and 2 clinical trials, population PK modeling and simulation, and phase 3 clinical trials of select patients and clinical scenarios designed for regulatory approval. There are vast differences between clinical trial populations and real-world patients in severity of illness [8], host physiology [9–12], and use of extracorporeal organ support treatments—all of which may lead to variable PK [10, 13–35]. Clinicians must extrapolate approved dosing

TABLE 1 | Summary of questions and consensus statements/recommendations for beta-lactam dose individualization.

Question	Statements and recommendations
What is the impact of beta-lactam dose individualization on patient-centered treatment outcomes?	Recommendation. The panel suggests the use of beta-lactam antibiotic dose individualization to improve patient-centered treatment outcomes (level of evidence: low, strength: conditional).
What PK/PD targets are preferred for beta-lactam individualization to improve treatment outcomes?	Recommendation. The panel suggests beta-lactam dose individualization programs pursue a minimum PK/PD target of 100% $fT_{>MIC}$ to improve the likelihood of clinical cure (level: very low, strength: conditional). The panel cannot recommend for or against specific targets for mortality, microbiologic cure, or length of stay (Level: low, Strength: conditional). Best Practice Statement. The panel recognizes the importance of loading doses and/or prolonged infusions to achieve a minimum PK/PD target of at least 100% $fT_{>MIC}$ with beta-lactam dose individualization. In infections at sites with decreased drug distribution (e.g., meningitis), a higher plasma PK/PD target such as 100% $fT_{>4 \times MIC}$ may be appropriate to ensure adequate drug concentrations at the target site.
What is the impact of beta-lactam dose individualization on patient-centered safety outcomes?	Recommendation. The panel cannot make a recommendation on the role for beta-lactam dose individualization to improve safety outcomes (level: low, strength: conditional). Best Practice Statement. Monitor for adverse events, particularly neurotoxicity, when beta-lactam dose individualization is performed.
What PK targets are preferred for beta-lactam dose individualization to improve safety?	Recommendation. The panel cannot make a recommendation on a specific PK target for beta-lactam dose individualization to improve safety (level: very low, strength: conditional). Best Practice Statement. It is the opinion of the panel that patients with acute or chronic kidney dysfunction, elderly patients, those expected to be on prolonged courses of therapy or at the extremes of body weight, and those at risk for drug accumulation, particularly when treated with beta-lactams such as cefepime (which has been frequently associated with neurotoxicity when these other risk factors are present), benefit from close monitoring of beta-lactam PK to prevent accumulation and mitigate toxicity risk.
For which patient populations should individualized beta-lactam dosing be considered?	Best Practice Statement. The panel supports the use of beta-lactam dose individualization in patients with critical illness (e.g., intensive care unit, sepsis, septic shock), use of extracorporeal devices including extracorporeal membrane oxygenation, kidney replacement therapy, plasmapheresis, and blood purification/adsorptive systems (e.g., hemoadsorption) and/or in patients with kidney dysfunction (e.g., acute kidney injury, chronic kidney disease) or augmented renal clearance. Beta-lactam dose individualization may be considered in other patient populations and treatment scenarios even when there is less evidence. For example, patients with pathophysiology that likely results in PK abnormalities or with difficult-to-treat infections may be candidates for beta-lactam dose individualization, including, but not limited to: hypoalbuminemia, infections due to drug-resistant organisms, clinical failure/insufficient antibiotic response, prolonged courses of antibiotic therapy, suspicion of toxicity, extremes of body weight (high or low), hypermetabolic conditions (e.g., cystic fibrosis), cancer, liver failure/cirrhosis, severe burn injury, pregnancy, and pediatric and neonatal patients.
Who are the key stakeholders and what is required to promote implementation of beta-lactam individualization strategies?	Best Practice Statement. Successful development and implementation of a beta-lactam dose individualization program requires a team of multidisciplinary stakeholders. The panel supports an implementation team with representation from institutional leadership, healthcare professionals, operational partners, regulatory support, and patients or their representatives. Stakeholder engagement is essential throughout program development and implementation, but involvement of specific individuals or groups at any given step of the process is guided by institution-specific need.

(Continues)

TABLE 1 | (Continued)

Statements and recommendations	
What are the main considerations involved with integrating beta-lactam dose individualization into the clinical workflow?	Best Practice Statement. The optimal method for integration of beta-lactam dose individualization into the clinical workflow depends on clinical practice structure and available resources. The panel supports patterning beta-lactam dose individualization workflows after those of other antibiotics such as vancomycin. After designing and implementing a beta-lactam dose individualization program into the clinical workflow, the panel supports iterative program evaluation at regular intervals.
What is the preferred specimen type for beta-lactam dose individualization?	Best Practice Statement. The panel supports the use of plasma or serum as the preferred specimen type for beta-lactam dose individualization.
What are the considerations for laboratory assay reporting in beta-lactam dose individualization programs?	Best Practice Statement. Clinical laboratories should follow established best practices and use a standard operating procedure for assay development, validation, calibration, and reporting. The panel suggests creation of a reporting plan and process including establishing interpretive criteria for values within, above, or below the typical range to support clinical decision making.
When multiple beta-lactam PK models exist for use in MIPD, what considerations should guide model selection?	Best Practice Statement. The panel supports the use of proactive MIPD with beta-lactam concentrations over traditional TDM, the latter of which is reactive. The panel recognizes that either approach is preferred to dosing without the guidance of individual concentration data and that access to MIPD is limited. The ideal population PK model for use in MIPD would be developed and validated in the patient population in which it will eventually be applied. If not possible, MIPD model selection should be guided by other factors including study population, sample size, covariates, sample density and validation methods.
What approach should be taken when there is a discordance between model predicted exposures and observed beta-lactam concentrations?	Best Practice Statement. In MIPD, the panel supports the use of a systematic approach to investigate discordances between observed concentrations and model-predicted concentrations before dosing changes are made. The assessment should include model appropriateness, accuracy of inputs (e.g., dose and concentration times), and stability of patient physiology/clinical status.
How should pathogen MIC be considered in dose individualization?	Best Practice Statement. The panel supports the use of a laboratory-defined MIC for beta-lactam dose individualization. If the reported MIC is within the susceptible range, the panel recommends the use of the susceptibility breakpoint or the epidemiologic cutoff value (ECV or ECOFF) (whichever is a higher numerical value) to guide beta-lactam dose individualization, rather than the isolate's reported MIC. When the isolate's MIC exceeds the susceptibility breakpoint or ECV/ECOFF, effectiveness and safety of other available antibiotics or antibiotic combinations should be considered. If no other safe antibiotic exists to which the organism tests susceptible, consideration could be given to use of one doubling dilution above the isolate MIC for beta-lactam dose individualization, provided the doses used to reach this threshold are safe in humans.
What are the relevant considerations for incorporating free (unbound) concentrations into beta-lactam individualization?	Best Practice Statement. Where free beta-lactam concentrations are available and validated, these are preferred. The panel notes free beta-lactam concentration assays are more technically complex, expensive, and not widely available. When measurement of free beta-lactam concentrations is unavailable, the panel supports estimating free concentrations from total concentrations using published protein binding values from the literature.

Abbreviations: $100\% fT_{>4\%MIC}$, Free drug concentration above 4 times the minimum inhibitory concentration of the organism for 100% of the dosing interval; $100\% fT_{>MIC}$, Free drug concentration above the minimum inhibitory concentration of the organism for 100% of the dosing interval; ECV/ECOFF, Epidemiologic cutoff value; MIC, Minimum inhibitory concentration [of the organism]; MIPD, Model informed personalized dosing; PK/PD, Pharmacokinetic/pharmacodynamic; TDM, Therapeutic drug monitoring.

TABLE 2 | Important terms and definitions.

Key term	Definition
Minimal inhibitory concentration (MIC)	Lowest concentration of a given antimicrobial that inhibits visible growth of an organism when tested in a standardized in vitro setting. The MIC is used to define antibiotic potency during drug development, to assess preclinical PK/PD relationships and when combined with clinical breakpoints, and to select antibiotics for a given patient.
Clinical (i.e., susceptibility) breakpoints	MIC cut-off value used to identify an isolate as susceptible, intermediate, susceptible dose-dependent, or resistant to a given antibiotic using typical doses; used for antibiotic selection for an individual patient
Epidemiological cutoff value (ECV, ECOFF)	The highest MIC of isolates without phenotypic resistance ('wild type')
100% $fT_{>MIC}$	Free drug concentration above the MIC of the organism for 100% of the dosing interval
100% $fT_{>4 \times MIC}$	Free drug concentration above 4 times the MIC of the organism for 100% of the dosing interval
Therapeutic drug monitoring (TDM)	Measurement of drug concentration in blood or other biological fluids for the purpose of individualizing therapy. Typically occurs on day 1–3 of therapy or after. May or may not be combined with the use of an up-front dosing model (i.e., population pharmacokinetic model, nomogram to aid initial dose selection). One or two specimens are collected to estimate the $fT_{>MIC}$.
Model-informed personalized dosing (MIPD)	Use of a population pharmacokinetic model to guide upfront dosing of beta-lactam therapy. This approach considers more covariates, in complex relationships, than would otherwise be used in traditional simplified dosing schema. Individual observed concentrations obtained at any time, combined with a population pharmacokinetic model, are then used to determine individual pharmacokinetic parameters and probability of pharmacokinetic/pharmacodynamic target attainment. Also known as model-informed <i>precision</i> dosing, but the panel favors the use of model-informed <i>personalized</i> dosing to balance prioritization of both accuracy and precision.
Beta-lactam dose individualization	Dosing regimen for one patient based on measured concentrations from that patient
Prolonged infusion beta-lactam therapy	Drug delivery over at least 2 h which broadly includes extended (non-continuous) and continuous infusion (over 24 h) therapy; use of prolonged infusions is a key tactic to achieve the minimum PK/PD target with beta-lactam dose individualization.
Beta-lactam concentration discordance	Numerical difference between the observed concentration measured in the patient and the concentration predicted by the model. Rarely will these numbers be in total agreement, i.e., that observed minus predicted = 0. Discordance is graded and must be evaluated in the context of the individual patient, but an absolute value of > 20% prediction error from the observation represents a threshold of concern based on extrapolation of FDA bioequivalence criteria [40]. The higher the discordance beyond this threshold, the less reliable the MIPD becomes.
Estimated free beta-lactam concentration	Estimated free concentration = total measured concentration multiplied by the fraction unbound

guidelines to these real-world patients with the goal of designing a dosing regimen that maximizes the probability of infection eradication and minimizes the risk of toxicity.

More than 400 population PK models have been published across 45 beta-lactam antibiotics which address beta-lactam PK. These data show wide variability in the estimated mean/median population parameter values (e.g., volume of distribution, clearance) [36, 37]. At the individual patient level, variation in beta-lactam PK parameter values can be even greater than at the population level (e.g., 10–30-fold) [38]. The observed variability in beta-lactam concentrations across real-world patients has prompted contemporary international clinical guidelines for infectious syndromes (e.g., sepsis, pneumonia) to encourage beta-lactam dose individualization [3–5, 39] to optimize drug effectiveness and safety. For the purposes of this consensus guidance, *beta-lactam dose individualization* is defined as a dosing regimen for one patient informed by measured concentrations from that patient (Table 2). Several

approaches to achieve beta-lactam dose individualization have been reported including traditional therapeutic drug monitoring (TDM), Model-Informed Personalized [Precision] Dosing (MIPD) using Bayesian estimation and other methods, nomograms, and decision trees [41–43]. Briefly, in traditional TDM, initial (“up front”) dosing is typically based on standardized guidelines or regulatory agency-approved population-based dosing schemes. One or more drug concentrations are then measured usually at least 1–3 days into therapy once steady state is reached to assess whether the patient's exposure is appropriate to achieve their clinical goals. Non-target concentrations would trigger dose adjustments, which are typically based on clinical intuition, experience, or simplified rules (e.g., dose scaling based on linear kinetics). In contrast, initial MIPD dosing is guided by somewhat more complex population models that incorporate patient characteristics (“covariates”) such as weight or organ function, and dosing is revised using concentrations with a Bayesian approach. Unlike traditional TDM, MIPD is designed to be proactive from the start of therapy,

aiming to achieve an exact target (i.e., rather than a range). It offers evidence-based dosing guidance to improve both accuracy and consistency in achieving desired drug exposures. The importance of both accuracy and precision (repeatability) prompts us to favor ‘Personalized’ over the frequently used “Precision” in “MIPD” for the purposes of this consensus document.

Clinical application of TDM for beta-lactams (and more recently MIPD) was first reported nearly 15 years ago, but comprehensive guidance on the key clinical, laboratory, and practical considerations associated with beta-lactam dose individualization techniques is lacking. There is a critical, unmet need to define the impact of beta-lactam dose individualization on patient outcomes, the minimum technical and clinical resources needed to support a dose individualization program, and the best implementation approach for such a program in a clinical setting.

1.1 | Scope and Purpose

The purpose of this document is to provide evidence-based recommendations on beta-lactam dose individualization. The target audience for these recommendations includes healthcare professionals caring for patients treated with beta-lactam therapy, i.e., practitioners in infectious diseases, critical care, internal medicine, pediatrics, pharmacy and pharmacology, and laboratory medicine. To determine the scope of these consensus recommendations, the panel considered randomized clinical trials [44–46], meta-analyses [47, 48], and guidelines or consensus statements [4, 39, 49]. The consensus group agreed that fundamentally, beta-lactam dose individualization requires the use and interpretation of measured beta-lactam concentrations in the patient to guide dose selection. Without measured concentrations, the panel considered the therapy to be population-based and not truly individualized, even if patient-specific characteristics (e.g., age, weight, creatinine clearance) were used to guide dose selection, due to the variability in observed PK across patients [25, 38]. The group acknowledged that prolonged infusions are one tactic that can be used to personalize beta-lactam dosing.

The scope of these consensus recommendations includes hospitalized patients across the age spectrum, receiving systemic beta-lactam antibiotics whether administered enterally or parenterally, regardless of infusion rate. The focus includes characterizing beta-lactam PK variability, PK/PD target selection, PK/PD optimization, clinical outcomes, and implementation approaches. The recommendations were developed through a rigorous review of the available literature, supplemented by expert guidance/input where studies were conflicting or lacking data, and a balanced assessment of the strengths and limitations of alternative approaches.

2 | Methodology

2.1 | Sponsorship and Endorsements

Pharmacotherapy Publications, Inc. (now the ACCP Foundation, Ltd.) supported the development of this consensus statement. Prior to any work being completed, seven societies/organizations with interest and expertise relevant to beta-lactam

individualization were identified, engaged, and provided feedback on the proposed methods. These endorsing groups included the American College of Clinical Pharmacy, European Society of Clinical Microbiology and Infectious Diseases, European Society of Clinical Microbiology and Infectious Diseases PK/PD of Anti-Infectives Study Group, Infectious Diseases Society of America, International Association of Therapeutic Drug Monitoring and Clinical Toxicology, Society of Critical Care Medicine, and the Society of Infectious Diseases Pharmacists. Endorsing organizations received updates on progress at intermittent intervals throughout statement development, reviewed the draft final consensus statement, were offered the opportunity to provide feedback, which was addressed in detail by the panel, and endorsed the final submitted version.

2.2 | Panel Selection

Two co-chairs were selected by Pharmacotherapy Publications Inc. The co-chairs assembled a panel of 19 subject matter experts in infectious diseases, critical care, pharmacy, laboratory medicine (inclusive of clinical chemistry and clinical microbiology), adult and pediatric practices, pharmacometrics, and guideline methodology. Candidate panelists were reviewed and approved by the Pharmacotherapy Publications Inc. Board. Empaneled members represented the sponsoring organizations and included a balance of expertise, gender, race/ethnicity, and geographic dispersion. Three consultants supported the panel including a patient representative, a research librarian, and a biostatistician. The panel was divided into workgroups, each with 3–6 individuals, one of which was the designated workgroup chair, that focused on four key domains (1) outcomes, (2) implementation, (3) laboratory considerations, and (4) modeling. Each workgroup was developed to ensure relevant expertise and diversity.

2.3 | Conflict of Interest Disclosure and Management

Candidate panelists self-reported any actual, potential, or perceived financial or intellectual conflicts of interest using a standardized disclosure form from the International Committee of Medical Journal Editors [50] (Appendix S1). All disclosures were reviewed by the co-chairs and the Pharmacotherapy Publications Inc. Board prior to any work being completed. No funding (or influence) was derived from the healthcare industry. Unresolvable conflicts of interest would have disqualified participation in the panel, but no conflicts of interest met this criterion. Workgroup assignments were made to minimize significant potential conflicts of interest. If a conflicts of interest was deemed significant, but not disqualifying, individuals recused from certain aspects of the consensus statement development including voting on recommendations relevant to the potential conflict. Every 6 months, panelists disclosed any new conflicts of interest which were reviewed by co-chairs.

2.4 | Development of Key Questions

A preliminary list of relevant questions for the consensus statement was created by the co-chairs and presented to the

panel for review and discussion. The consensus statements were structured as answers to a series of key questions [51, 52]. Each workgroup refined and prioritized candidate questions, which were then presented to the full panel through a series of virtual meetings. Feedback provided was used to make iterative refinements to the questions. The panel, including the patient representative consultant, reviewed the full list of questions for consistency and completeness, and a voice vote approved the list and progression to the literature evaluation phase. Ultimately, thirteen key questions framed the consensus guidance.

2.5 | Literature Search and Evidence Review

A systematic search of the literature was designed and performed by the consulting research librarian (R.A.) and peer-reviewed by a second research librarian. Briefly, the literature search included all relevant literature pertaining to the topics of *beta-lactam antibiotics AND individualization* found in Medline (Ovid), Embase (Elsevier), Web of Science (Clarivate), Cochrane Central Register of Controlled Trials (EBSCOhost), International Pharmaceutical Abstracts (EBSCOhost), [ClinicalTrials.gov](https://www.clinicaltrials.gov), WHO's International Clinical Trials Registry Platform, and the International Standard Randomized Controlled Trial Number Registry. The search strategy included articles in any language. Articles focused on animal models of infection or in vitro techniques, bioequivalence, or case reports were systematically excluded. Trials comparing prolonged infusion beta-lactam therapy to standard infusion beta-lactam therapy were excluded if no evidence of dose individualization was performed or assessed (e.g., beta-lactam concentration, PK/PD target attainment). Detailed literature search and evidence review are provided in Appendix S2. The original search was performed on September 1, 2023 and updated July 24, 2024. Title/abstract and full-text screening of studies were performed independently and in duplicate by at least two panel members using Covidence, a web-based software that supports evidence synthesis. Data extraction was also conducted independently and in duplicate using standardized forms. Co-chairs adjudicated all discrepancies.

2.6 | Development of Consensus Recommendations

Workgroup members extracted data from full-text articles to answer the identified key questions across three broad themes:

- Theme 1—What is the clinical impact of beta-lactam dose individualization?
- Theme 2—What are the minimum requirements for successful implementation of beta-lactam dose individualization?
- Theme 3—What are the technical and methodological considerations necessary to achieve beta-lactam dose individualization?

Individuals were asked to review and evaluate the identified literature, develop the recommendations, and summarize the evidence supporting each recommendation. Where possible, recommendations were informed by the results of systematic reviews of the literature and meta-analyses (detailed methods outlined in Appendix S3). A methodologic protocol was adhered to and analyses were performed in accordance with the PRISMA guidelines [53]. A detailed overview of the evidence review and consensus building process is outlined in Appendix S2. Briefly, for questions with sufficient high-quality evidence, the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach was used for evidence evaluation (Table 3) [54, 55]. Quality of evidence was assessed using ROB2 for randomized controlled trials [56] and the Newcastle-Ottawa Scale (NOS) for observational studies [57]. Strength and direction of GRADE recommendations also considered the evidence-to-decision framework [55]. For gradeable recommendations, the language of “the panel recommends” was used for strong recommendations, and “the panel suggests” was used for conditional recommendations. When high-quality evidence was insufficient, questions were addressed using best practice statements based on a consensus of the panelists. By definition, these ungradable best practice statements did not include a determination of recommendation strength or certainty of evidence. The full set of consensus recommendations and best-practice statements, along with the accompanying rationale, was drafted, iteratively refined, and approved by all panel members. The final version was circulated to sponsoring and endorsing organizations for final approval.

The final recommendations and best practice statements (Table 1) are intended to provide guidance for clinicians caring for hospitalized patients treated with beta-lactam therapy and those interested in implementing or refining a beta-lactam dose individualization program. These recommendations should not supersede clinical judgment for an individual patient but rather serve as a guide to practice.

TABLE 3 | Strength of evidence and recommendations [54, 55].

Term	Definition	Language
Level of Evidence	Underlying quality of supporting studies (randomized controlled trials, observational, risk of bias)	High/Moderate/Low/Very Low
Certainty of Evidence	Confidence that the true effect lies close to the estimate	High/Moderate/Low/Very Low
Strength of recommendation	Degree to which the panel is confident the benefits outweigh the harms	Strong ('the panel recommends') Conditional (sometimes referred to as weak; 'the panel suggests')

3 | What Is the Clinical Impact of Beta-Lactam Dose Individualization?

3.1 | Question 1. What Is the Impact of Beta-Lactam Dose Individualization on Patient-Centered Treatment Outcomes?

3.1.1 | Recommendation

The panel suggests the use of beta-lactam antibiotic dose individualization to improve patient-centered treatment outcomes (level of evidence: low, strength: conditional/weak).

3.1.2 | Rationale

A total of 11 studies informed this question, including 4 RCTs [44, 45, 58, 59] and 7 observational cohort studies [60–66]. Studies implemented beta-lactam individualization in a variety of ways and assessed a range of treatment outcomes including mortality, clinical cure, microbiological cure, hospital length of stay, and ICU length of stay. Outcome definitions varied across studies but similar outcomes were combined for analyses when medically justified. The panel reviewed these outcomes individually and in aggregate. Results are summarized below and in Appendix S4, Tables S1–S5. Additional information about key patient populations suitable for individualized beta-lactam dosing is expanded upon in Question 5.

3.1.2.1 | Mortality. Eleven studies reported mortality data, 7 of which were observational [60–66] and 4 of which were RCTs [44, 45, 58, 59]. Six of these studies were considered to be high quality [44, 58, 60, 61, 63, 64]. The overall pooled estimate from the random-effects frequentist meta-analysis showed that beta-lactam dose individualization had a non-significant decreased risk of mortality (OR 0.76 [95% CI 0.54–1.06]). Among RCTs only, the effect was more modest (OR 0.91 [95% CI 0.62–1.33]). Bayesian modeling yielded posterior probabilities of benefit of 97% under skeptical priors and 99% under neutral priors across all studies. The posterior probability of benefit was similar among studies of patients with a high severity of illness, but the probability of benefit was attenuated when limiting to higher quality studies (77%–93%) and to RCTs (61%–73%). Inconsistency in the observed effect led the panel to rate the certainty for this outcome as low.

3.1.2.2 | Clinical Cure. Nine studies, three of which were RCTs [44, 58, 59], evaluated clinical cure [44, 58–61, 63–66]. The pooled frequentist random-effects meta-analysis estimate indicated that beta-lactam individualization significantly increased the probability of clinical cure (OR 1.99 [95% CI 1.54–2.56]). This finding was consistent across subgroup and sensitivity analyses. The Bayesian posterior probabilities of benefit were as large as >99% overall and ranged from 71% to 98% (under skeptical and neutral priors) in RCTs [44, 58, 59]. Heterogeneity in clinical cure definitions led the panel to rate the certainty for this outcome as low.

3.1.2.3 | Microbiologic Cure. Five studies, two of which were RCTs [44, 58], evaluated microbiological cure [44, 58, 60, 63, 65]. The pooled frequentist random-effects meta-analysis

estimate suggested a non-significant increase in microbiologic cure (OR 1.42 [95% CI 0.73–2.76]). Heterogeneity in outcome definitions and imprecision in effect estimates led the panel to rate the certainty for this outcome as very low.

3.1.2.4 | Length of Stay. Hospital length of stay was reported in 7 studies [44, 59, 60, 62, 63, 65, 66], 2 of which were RCTs [44, 59], and ICU length of stay in five studies [44, 45, 60, 63, 65], 2 of which were RCTs [44, 45]. Beta-lactam dose individualization did not reduce hospital or ICU length of stay, with posterior probabilities of reducing hospital length of stay ranging from 48% to 59% and posterior probabilities of reducing ICU length of stay ranging from 52% to 59%. Imprecision in effects led the panel to rate the certainty for these outcomes as very low.

Overall, the panel recognized that there were few RCTs with data available to evaluate the impact of beta-lactam dose individualization on treatment outcomes and that this is an opportunity for future study. Depending on study quality, subpopulations evaluated, and prior beliefs about the credible effect sizes, the statistical significance for outcomes varied. Nevertheless, the panel observed a signal of benefit which consistently favored beta-lactam dose individualization particularly with respect to clinical cure. The evaluated study endpoints such as mortality and clinical cure were considered highly relevant to patients and there was no appreciable evidence of harm associated with performing beta-lactam dose individualization. A direct evaluation of the resource implications of routine beta-lactam individualization was not performed. The panel also recognized that the availability of and experience with beta-lactam dose individualization may vary. In particular, limited access to MIPD dosing software and drug assays with rapid turnaround times, although increasingly common worldwide [67], may limit the clinical benefit of dose individualization in some settings. Collectively, these findings informed the panel's conditional (weak) recommendation in favor of beta-lactam dose individualization, recognizing both signals of potential benefit and important limitations in the evidence base.

3.2 | Question 2. What PK/PD Targets Are Preferred for Beta-Lactam Individualization to Improve Treatment Outcomes?

3.2.1 | Recommendation

The panel suggests beta-lactam dose individualization programs pursue a minimum PK/PD target of 100% $fT_{>MIC}$ to improve the likelihood of clinical cure (level: very low, strength: conditional). The panel cannot recommend for or against specific targets for mortality, microbiologic cure, or length of stay (Level: low, Strength: conditional).

3.2.2 | Best Practice Statement

The panel recognizes the importance of loading doses and/or prolonged infusions to achieve a minimum PK/PD target of at least 100% $fT_{>MIC}$ with beta-lactam dose individualization. In infections at sites with decreased drug distribution (e.g., meningitis), a higher plasma PK/PD target such as 100% $fT_{>4 \times MIC}$ may be appropriate to ensure adequate drug concentrations at the target site.

3.2.3 | Rationale

Sixteen studies [9, 44, 68–81] addressed the relationship between PK/PD target choice and treatment outcomes with beta-lactam dose individualization, of which only two were RCTs [44, 73]. All but four studies [70, 78–80] included exclusively critically ill patients. Beta-lactam antibiotics were administered as a continuous infusion (over 24h) in most studies [9, 44, 69, 71, 72, 74, 76, 77] or as an extended infusion (over at least 2h) [70, 75, 78]. Preferred PK/PD targets for continuous infusion therapy may be higher [82] than with intermittent infusion; thus aggregation of data across studies may have introduced heterogeneity. While not all studies were specifically designed to compare treatment outcomes between those who achieved PK/PD targets and those who did not, sufficient data were available to manually tabulate differences. PK/PD targets differed across studies with most using 100% $fT_{>MIC}$ [68, 69, 73, 78–81], few using 100% $fT_{>4 \times MIC}$ [9, 74] and the remainder selecting alternative trough [70] or steady state [44, 71, 72] concentrations, or concentration to MIC ratios [75–77] (Table 2). The treatment outcomes evaluated in these studies included mortality [44, 68, 71, 73–75, 78, 80, 81], clinical cure [9, 69, 70, 72, 74, 78, 80, 81], microbiological cure [74, 76, 77, 79], and hospital [73, 78, 80] or ICU length of stay [68, 73, 81]. Results for each outcome are summarized below and in Appendix S4, Table S6. Effect estimates below are presented with 95% confidence intervals for frequentist analyses, and Bayesian analyses are reported using posterior probabilities relative to a null effect, where applicable.

3.2.3.1 | Mortality. Nine studies [44, 68, 71, 73–75, 78, 80, 81] reported on PK/PD target choice and mortality, five of which were considered high quality [44, 68, 71, 73, 81], two of which were RCTs [44, 73]. Collectively, the OR for mortality in patients achieving PK/PD targets versus those not achieving PK/PD targets was not significantly different (1.45 [95% CI 0.89–2.38]) irrespective of whether a standard (e.g., 100% $fT_{>MIC}$) or more aggressive target (e.g., 100% $fT_{>4 \times MIC}$) was used. The overall effect was similar when restricted to only high-quality studies (OR 1.55 [95% CI 0.88–2.73]). Therefore, the panel was unable to recommend for or against specific PK/PD targets for beta-lactam dose individualization to improve survival.

3.2.3.2 | Clinical Cure. Of the eight studies that assessed clinical cure in relation to PK/PD target attainment [9, 69, 70, 72, 74, 78, 80, 81], none were RCTs and only one retrospective cohort study [81] was of high-quality design. This retrospective cohort study by Wu et al. demonstrated a significant increase in clinical cure among 183 out of 225 (81.3%) critically ill patients who achieved 100% $fT_{>MIC}$ compared to the 130 out of 181 (71.8%) who did not (OR 1.71 [95% CI 1.07–2.72]). Though definitions of clinical cure were inconsistent across studies, three studies, including the article by Wu et al., observed a significant increase in clinical cure associated with PK/PD target attainment [9, 72, 81]. Most studies used a PK/PD target of 100% $fT_{>MIC}$ [69, 70, 78, 80, 81] and point estimates for all but two studies [70, 78] favored PK/PD attainment. The overall effect suggested that achieving study-defined PK/PD targets was not significantly associated with greater odds of clinical cure (OR of 2.16 [95% CI 0.89–5.29]). However, subgroup analyses showed significantly greater odds of clinical cure among prospective cohort studies (OR 3.1 [95% CI 1.28–7.52]) and those pursuing more aggressive

PK/PD targets (OR 8.63 [95% CI 3.24–22.98]). The panel considered the aggregate and sensitivity analyses, results from the higher quality cohort study [81], and the results of our analysis evaluating the impact of dose individualization on clinical cure (see Question 1). The panel determined the balance of data favored a minimum target of 100% $fT_{>MIC}$ to improve the likelihood of clinical cure (level: very low, strength: conditional).

3.2.3.3 | Microbiological Cure. The data for microbiological cure related to PK/PD target attainment are limited and inconsistent. Microbiological cure was reported in four observational studies [74, 76, 77, 79] and collectively showed significantly increased odds of microbiological cure associated with PK/PD target attainment (OR 16.15 [95% CI 3.29 to 79.38]). Most of these studies administered beta-lactam antibiotics as a continuous infusion and aimed for more aggressive PK/PD targets [74, 76, 77]. From our subgroup analysis by study design, the overall effect appears to have been driven by two retrospective cohort studies [76, 77] (OR 36.34 [95% CI 11.70 to 112.83]). The OR for microbiologic cure in the two prospective studies was 4.64 (95% CI 0.89 to 24.22) [74, 79]. The limited and inconsistent evidence, heterogeneity of study designs, and imprecise effect estimates with wide confidence intervals precluded the panel from making a recommendation for specific PK/PD targets to improve microbiologic cure. However, the panel acknowledges that optimizing beta-lactam exposure could improve the odds of achieving microbiologic cure.

3.2.3.4 | Length of Stay. Three studies reported on the relationship between PK/PD target attainment and hospital length of stay [73, 78, 80] and/or ICU length of stay [68, 73, 81]. The single RCT [73] found a favorable impact of PK/PD target attainment on hospital and ICU length of stay. In all except one study [78] beta-lactams were administered as a short infusion (e.g., ~30 min). Overall, random effects analysis found that achieving 100% $fT_{>MIC}$ was not associated with a significant difference in hospital length of stay [mean difference –4.63 (95% CI –9.82 to 0.56) days] or ICU length of stay [mean difference +1.01 (95% CI –3.71 to +5.74) days]. The limited evidence precluded the panel from rendering a recommendation for specific PK/PD targets for beta-lactam individualization to shorten length of stay.

Overall, while the data were heterogeneous, the panel concluded that achieving a minimum PK/PD target of 100% $fT_{>MIC}$ may improve the likelihood of clinical cure. However, the certainty of evidence was very low and there were insufficient data to support specific targets for other outcomes. Many studies assessing individualized dosing relied on assumed (e.g., MIC susceptibility breakpoints or ECV/ECOFF) rather than measured pathogen-specific MICs. This reliance can contribute to variability and imprecision in observed outcomes (see also the discussion in Question 12). While some studies suggested higher targets (100% $fT_{>4 \times MIC}$) may be beneficial, the panel emphasized that the doses necessary to consistently reach these higher targets in all patients would likely exceed safe licensed dosing practices. The panel noted that loading doses and prolonged infusions are often necessary to achieve the desired PK/PD target. Certain infection types (e.g., meningitis) may require higher exposure to achieve adequate concentrations at the site of infection and certain patient contexts may warrant tailored PKPD thresholds (e.g., septic shock). Doses higher than those approved by regulatory agencies may be necessary to achieve adequate drug exposure, but the

clinical appropriateness of this must be weighed on a case-by-case basis against other available alternatives.

3.3 | Question 3: What Is the Impact of Beta-Lactam Dose Individualization on Patient-Centered Safety Outcomes?

3.3.1 | Recommendation

The panel cannot make a recommendation on the role for beta-lactam dose individualization to improve safety outcomes (level: low, strength: conditional).

3.3.2 | Best Practice Statement

Monitor for adverse events, particularly neurotoxicity, when beta-lactam dose individualization is performed.

3.3.3 | Rationale

Nine studies comprised of four RCTs [44, 45, 59, 83] and five observational studies [60, 61, 63, 65] evaluated safety outcomes in the setting of beta-lactam dose individualization compared to standard of care. Studies that did not specifically evaluate dose individualization as an intervention or adverse events as an outcome were excluded from the analysis. Results are summarized hereafter and in Appendix S4, Table S7. The definitions for toxicity/safety endpoints varied across studies. Five studies evaluated the impact of beta-lactam dose individualization on patient safety without regard to baseline patient risk [44, 45, 59, 83, 84] (e.g., use of TDM to surveil drug concentrations for toxicity in allcomers including patients at low risk or without evidence of toxic features). Neurotoxicity was specifically assessed in six studies including two RCTs [45, 59] and four observational studies [60, 61, 63, 65]. The random-effects meta-analysis for neurotoxicity showed no significant difference between individualized dosing and standard of care, OR 0.94 (95% CI 0.45–2.00). No significant difference was noted between individualized dosing and standard care for nephrotoxicity in three observational studies [61, 63, 65] [pooled estimate of effect of OR 0.83 (95% CI 0.43–1.61)] or in the two observational studies [60, 65] that assessed hepatotoxicity [random-effects meta-analysis OR 0.90 (95% CI 0.53–1.54)]. In four other studies, including in three RCTs [44, 45, 83] and one observational study [63], safety was described as a composite outcome. The random-effects meta-analysis showed an OR 1.04 (95% CI 0.60–1.81) for beta-lactam dose individualization versus standard of care for the broad definition of any safety event. Taken together, point estimates generally favored improved safety with beta-lactam dose individualization versus standard of care for most of the adverse events reported. However, the panel had low confidence in the effect estimates given the relatively small numbers of studies evaluating adverse events, the indirectness of the evidence, heterogeneity in outcome definitions, and the variability in effect sizes. The application of beta-lactam individualization in consecutive patients without regard to baseline risk for adverse effects led to less evidence of a safety benefit [44, 45, 59, 83, 84] and the inability of the panel to render a recommendation on

the impact of beta-lactam dose individualization for safety outcomes. Practically however, beta-lactam individualization for the purpose of enhancing drug safety is typically evaluated on a per-patient basis [85] considering the baseline risk of adverse effects and the diagnostic suspicion of toxicity. The panel agreed that in this circumstance, beta-lactam dose individualization may be helpful in reducing risk (i.e., through initial dosing guided by MIPD) and detecting and mitigating adverse effects (i.e., through TDM). The panel suggests monitoring for treatment emergent adverse effects, especially neurotoxicity, in all patients treated with beta-lactam dose individualization as a best practice.

3.4 | Question 4: What PK Targets Are Preferred for Beta-Lactam Dose Individualization to Improve Safety?

3.4.1 | Recommendation

The panel cannot make a recommendation on a specific PK target for beta-lactam dose individualization to improve safety (level: very low, strength: conditional).

3.4.2 | Best Practice Statement

It is the opinion of the panel that patients with acute or chronic kidney dysfunction, elderly patients, those expected to be on prolonged courses of therapy or at the extremes of body weight, and those at risk for drug accumulation, particularly when treated with beta-lactams such as cefepime (which has been frequently associated with neurotoxicity when these other risk factors are present), benefit from close monitoring of beta-lactam PK to prevent accumulation and mitigate toxicity risk.

3.4.3 | Rationale

Five studies evaluated the relationship between PK/PD target choice and safety outcomes with beta-lactam dose individualization. These included three retrospective cohort studies [71, 86, 87], one prospective cohort study [88] and one RCT [44]. All studies included critically ill patients with sepsis or septic shock and evaluated the relationship between beta-lactam exposure and safety. Safety endpoints included mortality [44, 71, 86–88], neurotoxicity [71, 88], acute kidney injury [88], hepatic injury [88], total adverse events and serious adverse events [44]. Beta-lactams assessed included piperacillin-tazobactam [44, 86–88], meropenem [71], amoxicillin, cloxacillin, piperacillin, cefepime, ceftazidime, cefotaxime, ceftriaxone and imipenem [88]. Exposure definitions varied by study and included exposure ratios [44, 86] (e.g., C_{ss} 4xMIC) and absolute drug concentrations [71, 87, 88]. Data from these studies were not considered sufficient to meta-analyze; therefore, a narrative summary is provided with key findings in the results and further highlighted in Appendix S4, Table S8.

3.4.3.1 | Mortality. All five studies reported mortality [44, 71, 86–88], one of which was an RCT [44]. Three studies found higher mortality in patients with piperacillin steady state concentrations

≥ 96 mg/L [44, 86, 87], while Zerbib et al. reported a non-significant increase in ICU mortality in patients with beta-lactam $C_{\min}/MIC \geq 12.4$ [where MIC values used for calculation of the $C_{\min}$ (or C_{ss}) over MIC ratio were the EUCAST non-species-related PK/PD breakpoints] [88]. One study found no association between meropenem concentration and mortality after adjustment for severity scores [71]. While mortality was numerically higher in patients with elevated beta-lactam exposures in most studies, the relationship was not consistently adjusted for kidney dysfunction or illness severity, and the study designs precluded an assessment of causality. Likewise, the evaluated studies included different beta-lactams that have inherent differences in their PK properties.

3.4.3.2 | Neurotoxicity. Across studies in the broader literature, the panel recognized that cefepime is most commonly associated with concentration-dependent neurotoxicity in adults, with greater risk observed among patients with trough concentrations exceeding 20–35 mg/L [89, 90]. For piperacillin, meropenem and flucloxacillin, neurotoxicity has been reported at very high concentrations (e.g., $C_{\min}$ for piperacillin > 360 mg/L; meropenem > 64 mg/L, and flucloxacillin > 125 mg/L), but these safety thresholds are less well established [89]. During the systematic review process, the panel found that many studies relating PK exposures with toxicity outcomes were of lower quality and subject to potential biases. The panel therefore restricted the evaluation to higher quality studies (i.e., 7 or more stars on the NOS evaluation). Only two higher quality studies assessed the relationship between PK/PD targets and neurotoxicity [71, 88]. An exposure dependent relationship was found by Zerbib et al., with 75.7% (28 out of 38) of patients with excessive exposure, defined as $C_{\min}/MIC \geq 12.4$, experiencing neurological deterioration versus 33.3% (24 out of 81) below that threshold [88]. No such association was found in the other study [71].

3.4.3.3 | Other Safety Outcomes. A single high-quality observational study related beta-lactam PK thresholds with nephrotoxicity and hepatotoxicity outcomes. The study found trends toward greater nephrotoxicity and hepatotoxicity with higher drug exposures, though this was not statistically significant [88]. A single RCT found no difference in aggregated adverse events or serious adverse events with beta-lactam dose individualization (in this case TDM) versus standard of care [44]. The investigators found a single case of *Clostridioides difficile* disease attributable to the study drug in the TDM arm versus none in the control arm [44]. A single high-quality observational study evaluated the duration of mechanical ventilation, time to extubation, and time to ICU discharge, which were prolonged in patients with very high/excessive drug exposures [88]. Hematologic toxicity was not evaluated in this review.

Taken together, the data from these studies and the broader literature suggested to the panel that very high/excessive beta-lactam exposures may be associated with an increased risk of toxicity and mortality, but that clear toxicodynamic thresholds in acutely ill patients are uncertain. Limitations of existing literature include confounding by pre-existent kidney dysfunction, heterogeneous exposure definitions, lack of consistent and systematic safety assessment and lower quality of evidence. Among patients at risk of drug accumulation and with impaired drug clearance, the panel supports use of beta-lactam dose individualization to avoid excessive exposures (i.e., exposures in excess

of those needed to achieve maximal effectiveness benefit) and reduce the likelihood of toxicity.

3.5 | Question 5: For Which Patient Populations Should Individualized Beta-Lactam Dosing Be Considered?

3.5.1 | Best Practice Statement

The panel supports the use of beta-lactam dose individualization in patients with critical illness (e.g., intensive care unit, sepsis, septic shock), use of extracorporeal devices including extracorporeal membrane oxygenation, kidney replacement therapy, plasmapheresis, and blood purification/adsorptive systems (e.g., hemoadsorption) and/or in patients with kidney dysfunction (e.g., acute kidney injury, chronic kidney disease) or augmented renal clearance. Beta-lactam dose individualization may be considered in other patient populations and treatment scenarios even when there is less evidence. For example, patients with pathophysiology that likely results in PK abnormalities or with difficult-to-treat infections may be candidates for beta-lactam dose individualization, including, but not limited to: hypoalbuminemia, infections due to drug-resistant organisms, clinical failure/insufficient antibiotic response, prolonged courses of antibiotic therapy, suspicion of toxicity, extremes of body weight (high or low), hypermetabolic conditions (e.g., cystic fibrosis), cancer, liver failure/cirrhosis, severe burn injury, pregnancy, and pediatric and neonatal patients.

3.5.2 | Rationale

Evidence supports use of beta lactam dose individualization in patients with critical illness, kidney dysfunction, or augmented renal clearance (typically defined in adults as a creatinine clearance > 130 mL/min/1.73m²) [91, 92] (see also recommendations, statements, and evidence outlined for Questions 1–4) [93, 94]. There are numerous other populations and scenarios where observational studies and clinical reports have described the use of beta-lactam dose individualization, typically due to expected abnormalities in PK, altered host-response to infection, difficult to penetrate sites of infection, difficult to treat microorganisms, or heightened risk for adverse effects. In these scenarios, the motivation for beta-lactam dose individualization may be to optimize effectiveness (e.g., augmented renal clearance), safety (e.g., cefepime use in a patient with new encephalopathy), or both. The panel also acknowledged that beta-lactam dose individualization may be more appropriate for some beta-lactams than others.

3.5.3 | Future Directions on the Clinical Impact of Beta-Lactam Dose Individualization

Additional research is needed to clarify the relationship between beta-lactam dose individualization and outcomes and to define the optimal PK/PD targets. Although existing evidence suggests potential treatment benefits, the precision of current estimates remains low. Future studies of beta-lactam dose individualization should be multicenter, prospective, ideally randomized, and sufficiently powered to assess clinically meaningful outcomes

such as survival and clinical cure [46, 47]. Definitions for clinical endpoints such as cure should be transparent, well-justified, and rigorously evaluated. Studies should be of sufficient size and statistical power to evaluate subgroups and conduct sensitivity analyses to probe the heterogeneity of treatment effects according to baseline illness severity, infection source, pathogen type and antimicrobial susceptibility, body habitus, and end-organ function. MIC determination strategy should be standardized and transparently reported in these trials and PK/PD targets should be pre-specified. Enrichment techniques (i.e., predictive, prognostic) could be used to enroll only patients most likely to benefit from beta-lactam dose individualization. Most studies to date have focused on adult patients, but children/neonates present with unique PK/PD considerations that require separate evaluation (see also Special Considerations section). Studies need to disentangle MIPD from traditional TDM to isolate the aspect(s) of beta-lactam dose individualization most likely to be beneficial [95]. Recent trial designs have also led to delays in the introduction of the intervention (e.g., 1–3 days into therapy or longer). Alternative trial designs (e.g., cluster randomized, pragmatic) could improve timely allocation to the intervention, and minimize the overall resources required for the clinical trial thereby enhancing the potential to more efficiently recruit larger and more diverse samples.

Although the panel agrees that a minimum PK/PD target of $100\% fT_{>MIC}$ is recommended for treatment outcomes, optimal targets could vary based on specific clinical scenarios. For example, higher targets have been suggested for continuous infusion beta-lactam therapy (i.e., $100\% fT_{>4 \times MIC}$). Future trials will need to quantify clinical outcomes associated with different PK/PD targets (e.g., standard vs. more aggressive) [96] and contextualize the target according to the infusion type [49]. This is of particular interest in populations with deep seated or penetration-limited infections (e.g., CNS infections), where serum monitoring may not fully reflect the concentration at the site of infection, which may motivate more aggressive serum targets.

There is a need to define and validate exposure thresholds associated with beta-lactam toxicity, particularly for agents like cefepime and piperacillin in critically ill patients. Standardized, systematic assessment of adverse drug effects, including neurotoxicity, nephrotoxicity, hepatotoxicity, and hematologic toxicity, should be incorporated into study design. Randomized clinical trials or large prospective cohorts with TDM, illness severity adjustment and enhanced characterization of kidney clearance using novel biomarkers are needed to separate exposure-toxicity from confounding due to organ dysfunction and sepsis severity. Mechanistic studies exploring individual susceptibility to toxicity may help refine patient-specific safety targets. Integration of Bayesian pharmacometric frameworks that incorporate prior knowledge about susceptibility to adverse effects and real-time patient data may aid in safe dose individualization. International collaborative efforts to establish registries with standardized outcome reporting for both effectiveness and safety would aid in systematic gathering of evidence to refine and validate future beta-lactam dose individualization strategies.

Finally, this guidance and these specific questions are focused on patient-centered outcomes, but the ecological impact of beta-lactam dose individualization should also be considered. Future studies could explore the impact of beta-lactam dose

individualization on cutaneous, intestinal, and other microbiota (more specifically, the “resistome”) [97].

3.6 | Panel Perspective on the Clinical Impact of Beta-Lactam Dose Individualization

The panel acknowledged that beta-lactam dose individualization with use of TDM or MIPD is an evolving practice. Complexity, resource requirements, and opportunity costs associated with implementation of such programs informed the conditional (weak) nature of the recommendations favoring dose individualization to improve patient-centered outcomes, consistent with the a priori defined methodology plan using GRADE methodology and the Evidence to Decision (EtD) framework. Despite limitations in the evidence base, the systematic review and meta-analyses demonstrated evidence of benefit for patient-centered efficacy outcomes, particularly PK/PD target attainment and clinical cure, without consistent evidence of harm. While the panel was unable to make outcome-specific recommendations for safety, it agreed that excessive beta-lactam exposure is plausibly associated with toxicity in high-risk populations and that dose individualization may mitigate this risk in selected patients. The guidance is not intended to mandate universal implementation and recognizes that adoption depends on local resources and priorities (see also Questions 6 and 7 and the Special Considerations section devoted to resource-limited settings). The guidance instead provides a contemporary, rigorous appraisal of the existing evidence, a framework for centers that elect to develop these services, and an acknowledgment that additional research including formal economic evaluations is needed.

4 | What Are the Minimum Requirements for Successful Implementation of Beta-Lactam Dose Individualization?

4.1 | Question 6: Who Are the Key Stakeholders and What Is Required to Promote Implementation of Beta-Lactam Individualization Strategies?

4.1.1 | Best Practice Statement

Successful development and implementation of a beta-lactam dose individualization program requires a team of multidisciplinary stakeholders. The panel supports an implementation team with representation from institutional leadership, healthcare professionals, operational partners, regulatory support, and patients or their representatives. Stakeholder engagement is essential throughout program development and implementation, but involvement of specific individuals or groups at any given step of the process is guided by institution-specific need.

4.1.2 | Rationale

Implementation of a beta-lactam dose individuation program is complex and can be supported by application of theoretical frameworks grounded in implementation science, quality improvement, and behavior change. We refer the reader to previously published experience with application of these methods

to beta-lactam dose individualization program development and implementation [94, 98–102]. Early implementation requires members of the institution to internalize the importance of, make sense of, and agree to adopt the practice [99]. Support from institutional leadership ensures adequate resourcing and policy alignment. A return-on-investment analysis may be required to characterize the expected impact of a beta-lactam dose individualization program on patient-centered outcomes, acquisition cost of antibiotics, preparation and administration costs, personnel time, assay costs, and information technology

resources. Several studies have identified economic advantages of PK/PD- and/or TDM-based antibiotic prescribing attributed to decreased drug costs from reductions in total drug utilized per patient (in mg) [103, 104] or shorter durations of antibiotic therapy [105]. We are aware of only one study that assessed healthcare costs in critically ill patients treated with beta-lactam dose individualization [106]. A comprehensive evaluation of the health-economic impact of beta-lactam dose individualization is lacking. Early engagement of a multidisciplinary group of stakeholders as shown in Table 4 facilitates role delineation, optimal

TABLE 4 | Stakeholders involved with beta-lactam dose individualization and role(s).

Role	Stakeholders	Possible roles and contributions by stakeholder
Clinicians	Physicians and advanced practice providers including those with expertise in infectious diseases, critical care, transplant, and hematology-oncology. Colleagues in neurology and neurosurgery may have particular interest in beta lactam dose individualization for cefepime. Pharmacists and clinical pharmacologists including individuals with specialized training in infectious diseases, pharmacokinetics and pharmacodynamics, critical care, transplant, and hematology-oncology. Laboratory medicine (e.g., Microbiologists)	Involvement: During protocol development, training phases, and ongoing clinical outcome evaluations. Essential to develop clear, standardized guidelines and processes. Reasons: Providing clinical expertise, facilitating adoption, and monitoring for safety and effectiveness.
Leadership	Antimicrobial stewardship teams (co- led by pharmacists and physicians) Nursing Executive leadership (e.g., pharmacy and therapeutics committee, pharmacy administration, quality/regulatory, clinical teams for relevant patient units, health system or hospital administration) Finance Information Technology Laboratory medicine	Involvement: At the start for strategic planning and resource allocation, during policy approvals, and in critical evaluations. Reasons: Ensuring alignment with institutional objectives, approving key changes, facilitating clinician buy-in, and providing necessary resources.
Operational partners	Clinical laboratory and/or special chemistries laboratorians Microbiology laboratorians Supply chain Nursing Phlebotomists Electronic health record build including information technology Sterile products area technicians and pharmacists; robotics as applicable Communications staff	Involvement: In technical design, workflow optimization, and continuous technical support. Information technology teams will determine the ability to the order assay, build forcing functions in the electronic health record to guide appropriate use of the assay, create custom medication builds for specific doses, and integrate PK/PD software (if applicable). Supply chain representatives must ensure adequate supply of testing equipment. Reasons: Ensuring smooth integration into existing systems, optimizing processes, and resolving technical issues.
Patients	Patients Patient representatives Patient caregivers	Involvement: During the design phase, patient experience evaluation, and clinical impact review. Reasons: Offering crucial patient perspectives, providing feedback for improvement, and ensuring acceptance and adherence.
Regulatory Support	Institutional, state, or national policy	Involvement: Oversight regarding which healthcare providers can engage in various aspects of the program Reasons: Determine who can order, draw, transport, evaluate, and document patient specimens (and how and when these occur for different stakeholders)

lines of communication, and streamlined decision-making both during program development and throughout implementation [39, 98, 100, 107–110]. Pharmacists and/or pharmacologists are key operational stakeholders for beta-lactam dose individualization [98, 99]. In environments without access to clinical pharmacists or pharmacologists, dedicated team members including physicians, or other trained staff are primarily responsible for beta-lactam dose individualization. Nurses are essential to facilitate appropriate beta-lactam administration, sample collection for TDM, and patient monitoring toward therapeutic goals (effectiveness and safety). In addition to bedside clinicians, engagement of individuals from the clinical or special chemistry laboratory is needed to ensure access to an accurate and timely assay (preferably on site) that meets regulatory standards for clinical practice use [100]. Likewise, partners in the microbiology laboratory should agree to report the MIC for appropriate organisms to individualize beta-lactam dosing. Involvement of informaticists is needed to facilitate seamless technical integration into routine workflows. Finally, the panel emphasized that patient stakeholder(s) should be involved during the design phase of a beta-lactam dose individualization program to optimize the patient experience and to facilitate acceptance of individualized regimens, including if and how beta-lactam TDM results are reported to patients.

4.2 | Question 7: What Are the Main Considerations Involved With Integrating Beta-Lactam Dose Individualization Into the Clinical Workflow?

4.2.1 | Best Practice Statement

The optimal method for integration of beta-lactam dose individualization into the clinical workflow depends on clinical practice structure and available resources. The panel supports patterning beta-lactam dose individualization workflows after those of other antibiotics such as vancomycin. After designing and implementing a beta-lactam dose individualization program into the clinical workflow, the panel supports iterative program evaluation at regular intervals.

4.2.2 | Rationale

Beta-lactam dose individualization programs are relatively uncommon globally, but are more prevalent in Europe and Australasia than in the United States [67, 93, 111–115].

The development of unambiguous evidence-based guidelines and a standardized workflow for the ‘who’ and ‘how’ of beta-lactam dose individualization increases the likelihood of program success [98, 100, 110]. Implementation of beta-lactam dose individualization programs will require adaptation to local and national health-care structures, payment models, and workforce capacity. Where possible, procedures should be embedded in the electronic health record and aligned with legacy workflows for other local antibiotic individualization programs (e.g., vancomycin). Once a workflow is established, it is advisable to test the process in one or more pilot units/sites within the institution(s) to address gaps prior to widespread implementation (Appendix S5).

Patient selection for beta-lactam dose individualization is a key component of program design (see also Question 5). Some studies have described the use of beta-lactam TDM in select environments (e.g., ICU), in certain patient populations (e.g., burn patients) [116], or in a phased approach (e.g., starting in select ICU patients and expanding the program over time) [100]. The ideal candidates for beta-lactam dose individualization are those for whom drug monitoring is likely to improve patient-centered treatment and/or safety outcomes (see also recommendations and statements for Questions 1–5). To improve equity and access to specialized services, patients would ideally be proactively identified for dose individualization eligibility using clinical decision support rather than relying only on consultation or referral for this service.

Early in program development, use of MIPD or other up front dosing methods requires a needs assessment to understand the suitability of available PK models in these software packages and their fit for purpose, electronic health record integration, licensing costs, data privacy, and storage [117]. Programs must also determine the beta-lactam concentration sampling strategy and decide how many specimens are required per patient (based on type of dose individualization being performed and infusion type), when specimens should be taken, how frequently sampling can be repeated, and how to handle incorrectly collected specimens.

Assay considerations must be addressed during program development [102]. Notably, this consensus recommendation focuses on either serum or plasma concentration monitoring, which are generally treated as interchangeable in the literature [118]. Assays should be electronically orderable with results integrated into the patient’s medical record. During program development, it needs to be decided whether all providers will have access to beta-lactam concentration ordering or whether it will be restricted to select individuals (e.g., pharmacists or pharmacologists with or without physician co-signature). Regulatory alignment and local practice may dictate which team members can engage in which aspects of the program. A validated assay, performed locally, is preferred to facilitate rapid turnaround of results, ideally within the same day. When available, ‘in-house’ laboratories often perform beta-lactam assays in batches during regular business hours, rather than offering this testing continuously, which may limit utility [119]. Delays in results inherently reduce their actionability. Rapid turnaround testing is necessary for routine clinical application in hospitalized patients, especially the critically ill. Where local assays are unavailable, send-out assays (e.g., using reference laboratories) may be considered, but specimen timing, handling, and turnaround times greater than 1 day reduce the utility of the result for dynamic patients [99]. An ideal beta-lactam dose individualization workflow should articulate who collects the specimens, how they are delivered to the laboratory, and who is responsible for interpreting and acting on the results.

Effective results interpretation depends on having a clearly defined PK/PD target for the patient, a sense for the patient-specific risks of sub- and supratherapeutic exposure, and accurate documentation of the timing of medication administration [120, 121]. Institutions seeking to incorporate beta-lactam dose individualization into clinical workflows would ideally employ infusion pumps with interoperability that provide a two-way connection with the electronic health record and ensure pump libraries allow

programming of continuous infusions if utilized at the institution. The advantage of interoperability is that the infusion pump and electronic health record share and use data in real time, thereby preventing and identifying medication errors and thus improving the safety and accuracy of individualized dosing [122].

Qualitative studies indicate that sufficient, well-trained staff are needed for successful implementation of a beta-lactam dose individualization program. Stakeholders desire access to the underlying evidence, repeated exposure to key concepts, and access to experts with real world experience [98–100]. Education must be delivered at program inception and iteratively thereafter which may take the form of regular team meetings and case review/discussions for continuous quality improvement [98, 110]. Education programs should be tailored to the needs of different stakeholder groups. Forums beyond the classical ‘grand rounds’ style lecture are encouraged [98, 99].

Most often, beta-lactam dose individualization programs are run either as part of a consultative service (e.g., pharmacology service, PK service, infectious diseases consultation team) [94, 115] or by the primary clinical team (e.g., the critical care team), supported where possible by a clinical pharmacist or pharmacologist [98, 114, 119, 123, 124]. Competing priorities for these individuals in their primary roles must be balanced for the success of the beta-lactam dose individualization program. Notably, ongoing education for and performance of beta-lactam dose individualization is time consuming and often unpaid work. Clinicians report concerns related to increased workloads and have suggested that implementing a beta-lactam dose individualization program would require a formal arrangement for incorporation into their clinical practice [98]. Antimicrobial stewardship teams are a particularly well-suited group for evaluating program performance (e.g., protocol compliance, deviations, barriers, and provision of education) and these tasks should be integrated into stewardship dashboards to facilitate continuous quality improvement [125].

4.3 | Question 8: What Is the Preferred Specimen Type for Beta-Lactam Dose Individualization?

4.3.1 | Best Practice Statement

The panel supports the use of plasma or serum as the preferred specimen type for beta-lactam dose individualization.

4.3.2 | Rationale

Due to the relative ease of collection and interpretation, most published beta-lactam assays use plasma or serum, distinguished by the presence (plasma) or absence (serum) of clotting factors. There are only modest differences between these two specimen types for beta-lactam concentration evaluation [118]. The advantage of plasma is that it can be immediately processed, without waiting for blood to clot. Immediate processing allows for centrifugation, harvesting and processing/freezing, which is important for drugs like beta-lactams which are known to be unstable at room temperature [126, 127]. Use of plasma or serum is timely (showing what is happening at the moment of collection) and

it is proximal (blood delivers the drug to the target site, either directly or after diffusion into tissues). Whole blood specimens are relatively easy to collect compared to other specimen types and familiar to health professionals and clinical laboratories. Plasma or serum can be analyzed shortly after collection, or can be frozen for future processing. These features make it relatively inexpensive to work with, and in general, whole blood can easily be converted to the serum or plasma used for the assays. There are many assay types that could be suitable for processing plasma or serum samples. The most popular are based on high performance liquid chromatography (HPLC) combined with various detection methods (e.g., mass spectrometry). Although direct measurement of unbound beta-lactam concentrations provides the most accurate representation of pharmacologically active drug, it is more resource-intensive and costly to perform and not universally available. Given the current uncertainty regarding its incremental clinical benefit, total concentrations remain acceptable for most settings and most beta-lactams, provided that the protein binding characteristics of the drug are considered in the PK/PD target attainment interpretation (See also Question 13).

4.3.3 | Alternative Matrices

There are select situations in clinical practice when a more detailed characterization of beta-lactam disposition is needed to evaluate penetration to the site of infection. Urine, pulmonary fluid, and cerebrospinal fluid are examples of alternative matrices [93]. To measure drug in these fluids, validation must be performed since many assays have not been approved for non-blood specimens. Urine specimens may be appropriate for use as a specimen type for cystitis, but not appropriate for other infection types. Urine beta-lactam concentrations are affected by kidney function, hydration status, ureter obstruction, and instability of the beta-lactam while transiting the genitourinary tract. Pulmonary fluid and cerebrospinal fluid require more invasive procedures for collection than venipuncture and are collected for beta-lactam individualization only in specific circumstances (e.g., research studies). Tissue specimens can be challenging to collect and process, and tissue concentrations of antibiotics are subject to a large number of caveats that have been reviewed previously [128]. Non-blood matrices are less practical than plasma or serum, and thus the panel agreed that these alternatives are not preferred for routine clinical beta-lactam dose individualization. Dried blood spots and sampling of blood via newer volumetric sampling devices exist, but more data are needed before widespread clinical adoption is possible [129].

4.4 | Question 9: What Are the Considerations for Laboratory Assay Reporting in Beta-Lactam Dose Individualization Programs?

4.4.1 | Best Practice Statement

Clinical laboratories should follow established best practices and use a standard operating procedure for assay development, validation, calibration, and reporting. The panel suggests creation of a reporting plan and process including establishing interpretive criteria for values within, above, or below the typical range to support clinical decision making.

4.4.2 | Rationale

During assay development and validation, key considerations to be addressed include: the calibration range of the assay, how the calibration range compares to the range of possible clinical values, how results that are above or below the calibration range are approached, response to highly improbable results, quality control monitoring, and results presentation. Assay validation is frequently standardized by external regulatory bodies [130]. For example, the Food and Drug Administration (FDA) Guidance on Bioanalytical Methods [130] is a reasonable starting place for HPLC-based assays, along with an analysis of the checklist requirements from the College of American Pathologists (CAP). Most United States-based laboratories are inspected and certified by CAP, making compliance with their requirements essential. In other settings, external quality control programs can play a similar role in ensuring the quality of sample bioanalysis.

Procedures for results reporting should establish how results are presented to the clinicians, the “normal” values and any embedded clinical guidance. ‘Normal’ values (i.e., what usually should be expected) and “therapeutic range”, “target range”, or “target concentrations” (i.e., what concentrations are predictive of clinical outcomes) must be determined and presented to clinicians thoughtfully. A diverse group of stakeholders should be engaged when deciding the optimal local approach for results reporting (see also Questions 6 and 7) [94, 100]. Critical or semi-urgent values may need to be added to meet reporting requirements. All steps of the testing and reporting process must be included in a standard operating procedure.

4.5 | Future Directions for Implementation of Successful Beta-Lactam Dose Individualization

Future beta-lactam dose individualization studies must probe several key themes relevant to implementation. The scope of this consensus statement and the majority of existing published data focus on hospitalized patients, but individualization of beta-lactam therapy at care transitions (i.e., hospital to home) and in patients in the outpatient setting remains an opportunity [131]. The development and access to the technology needed to perform dose individualization remains a barrier; studies of methods to overcome this barrier are needed to promote optimal uptake. Exploration of key workflow considerations to enhance the relevance of send-out laboratory testing for patient care decision making, utilization of alternative assay methods in low-resource settings, real-time monitoring of drug concentrations, and development and validation of biosensing devices are opportunities for the future. Clinician workload has steadily increased due to the rising number and complexity of acutely ill patients, the influx of vast and multidimensional clinical data [132], escalating documentation demands, and growing expectations for participation in diverse non-clinical responsibilities [133]. Development of user-driven beta-lactam dose individualization clinical decision support systems is necessary to avoid further increasing the burden, and in turn, decreasing enthusiasm for uptake of this evidence-based practice. Health-economic evaluations are warranted to quantify the cost-effectiveness and broader potential system impact of beta-lactam dose individualization.

5 | What Are the Technical and Methodological Considerations Necessary to Achieve Beta-Lactam Dose Individualization?

5.1 | Question 10: When Multiple Beta-Lactam PK Models Exist for Use in MIPD, What Considerations Should Guide Model Selection?

5.1.1 | Best Practice Statement

The panel supports the use of proactive MIPD with beta-lactam concentrations over traditional TDM, the latter of which is reactive. The panel recognizes that either approach is preferred to dosing without the guidance of individual concentration data and that access to MIPD is limited. The ideal population PK model for use in MIPD would be developed and validated in the patient population in which it will eventually be applied. If not possible, MIPD model selection should be guided by other factors including study population, sample size, covariates, sample density, and validation methods.

5.1.2 | Rationale

While it is likely that no perfect model will exist to capture all the factors for each individual patient, it is the panel's opinion that imperfect models that are fit for purpose to guide dosing are better than the status quo, in which concentrations are not measured and models are not used; instead, doses are adjusted empirically. The panel agrees that the optimal approach is to adapt and validate an existing PK model for the local-context for each beta-lactam intended for use with MIPD. Should a suitable PK model not exist, development of a new population PK model, suitable for MIPD, that can be locally validated, is preferred. Population PK models developed for data characterization may not perform well when used for MIPD (i.e., Bayesian forecasting). Relevant methods should be considered in model development (e.g., leave-one-out, limited sampling analysis) to establish internal validity for MIPD. The model would ideally be able to incorporate specific demographic, clinical, and dosing factors unique to the local context, population, and drug. To develop such a model, the reader is directed to a previously published detailed overview of PK modeling methods [134, 135]. The panel recognized the significant challenges of developing a new PK model outside of large, resource-rich, academic environments, with access to pharmacometrics experts. In cases where a new PK model cannot be developed locally, use of published beta-lactam PK models available in commercial software is acceptable for MIPD. The panel reinforced that not every model will be appropriate for all patients. When evaluating existing beta-lactam PK models in MIPD software, key considerations include:

- Representativeness of the source population including the range of covariates assessed and the comparability to the population in which the model will be applied. For beta-lactams, the minimum essential covariates in the PK model should include a body size metric (e.g., weight, body mass index, body surface area), especially in children or patients at body size extremes, and a kidney function metric (e.g., estimated glomerular filtration rate).

- Duration of beta-lactam infusion (intermittent vs. prolonged vs. both).
- Number of patients and number and distribution of beta-lactam concentrations used to assess the accuracy and precision of the parameter estimates and the ability to estimate between subject variability.
- Evaluation of the model for MIPD versus descriptive PK modeling.

More details on model selection for MIPD can be found in a previously published tutorial [136]. Once selected, the preferred PK model should be validated with local data from the target population to determine the predictive performance (i.e., “external validation”). Validation may be performed by local experts or, more likely, in collaboration with MIPD software vendors. Of note, a distinct advantage of the MIPD framework is the ability to tailor dosing to the individual patient—ultimately enabling improvements in subpopulation predictions one patient at a time.

5.2 | Question 11: What Approach Should Be Taken When There Is a Discordance Between Model Predicted Exposures and Observed Beta-Lactam Concentrations?

5.2.1 | Best Practice Statement

In MIPD, the panel supports the use of a systematic approach to investigate discordances between observed concentrations and model-predicted concentrations before dosing changes are made. The assessment should include model appropriateness, accuracy of inputs (e.g., dose and concentration times), and stability of patient physiology/clinical status.

5.2.2 | Rationale

Once a model has been selected and applied for MIPD, one may encounter beta-lactam concentration discordance (Table 2), where model predicted exposures based on Bayesian estimation differ significantly from measured beta-lactam concentrations. Beta-lactam concentration discordance is variable, but an absolute value of >20% prediction error from the observation represents a threshold of concern [40]. Use of a systematic approach as shown in Figure 1 will guide the assessment of beta-lactam concentration discordance and next steps. In addition to issues with model appropriateness, errors in model inputs including doses, timing, covariates, concentration results, the stability of patient physiology, and the assay accuracy and precision must be considered. Specifically, it should be recognized that the measured concentration represents the best estimation of the “actual (i.e., nominal) concentration.” The FDA and European Medicines Agency indicate an allowable range for assay accuracy and precision of $\pm 15\%$ of the expected concentrations ($\pm 20\%$ at the lower limit of quantification) [130, 137]. Assay performance outside these bounds may explain beta-lactam concentration discordance. An understanding of the beta-lactam assay accuracy and precision, the model predicted beta-lactam concentration, and the relationship of model predictions to the actual,

measured concentration can be used to estimate the potential impact on the estimated $fT_{>MIC}$. These assessments should be made clear in software or as independently analyzed by experts in pharmacometrics.

If no clear explanation for biased predictions in MIPD can be identified and addressed, the panel supports repeating the concentration(s) using a new specimen as soon as possible, even if there has been an intervening dose change. With MIPD and Bayesian methods, levels can be interpreted without a need to wait for a new steady state for sampling. Such methods identify the most likely set of individual PK parameters that led to the observation(s) allowing forward and backward predictions. Gathering more than one specimen in the dosing interval may more clearly characterize the likely concentration-time profile. Most models in clinical use do not account for within-patient or inter-occasion random variability. Therefore, the model will seek to identify the most likely set of parameter values to cover the entire history of doses and measured concentrations with the least error. If the patient's clinical status is dynamic, or there is an interruption in beta-lactam therapy (e.g., drug discontinued and restarted shortly thereafter), it may be useful to only consider the most recent concentrations and dosing events in the Bayesian framework even if earlier data are available for the patient. Ultimately, the clinical team must decide in the context of the patient how to interpret poor model predictions.

If the investigation is unrevealing after applying the suggested approach, predictions from other models may be explored. This may require tools outside of the usual software application at the clinical team's disposal. Freely available online resources from reputable institutions may be considered. If accurate predictions are still not achieved, the panel recommends alternative approaches to assessing exposure that are not informed by models. For continuous infusion therapy, a concentration can be measured at any point after steady state is achieved, typically within 4–5 half-lives. For intermittent dosing, a peak and trough can be used to calculate the relative time after a dose at which the concentration will fall below the MIC.

5.3 | Question 12: How Should Pathogen MIC Be Considered in Dose Individualization?

5.3.1 | Best Practice Statement

The panel supports the use of a laboratory-defined MIC for beta-lactam dose individualization. If the reported MIC is within the susceptible range, the panel recommends the use of the susceptibility breakpoint or the epidemiologic cutoff value (ECV or ECOFF) (whichever is the higher numerical value) to guide beta-lactam dose individualization, rather than the isolate's reported MIC. When the isolate's MIC exceeds the susceptibility breakpoint or ECV/ECOFF, the effectiveness and safety of other available antibiotics or antibiotic combinations should be considered. If no other safe antibiotic exists to which the organism tests susceptible, consideration could be given to the use of one doubling dilution above the isolate MIC for beta-lactam dose individualization, provided the doses used to reach this threshold are safe in humans.

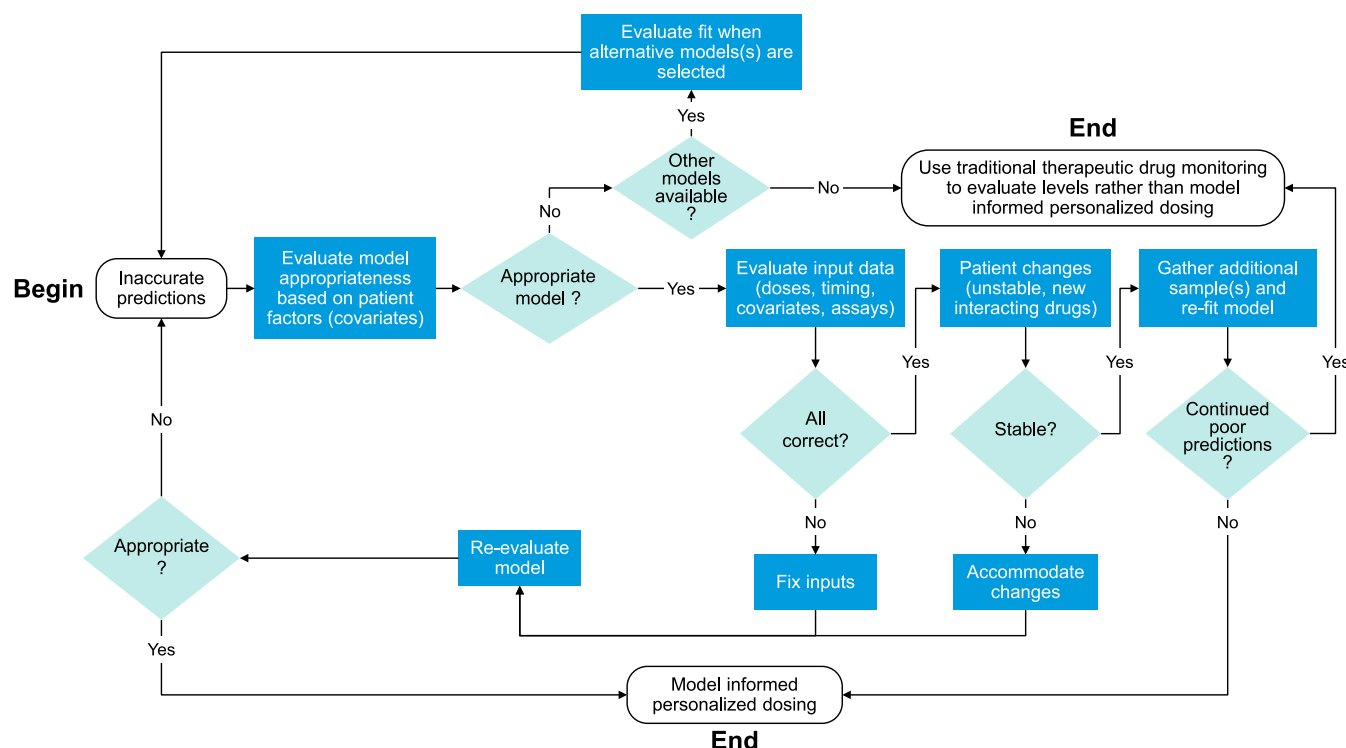


FIGURE 1 | Systematic approach to investigate discordances between observed beta-lactam concentrations and model-predicted concentrations. The first response to assess discordance between model-predicted concentrations and observed (i.e., measured) concentrations is to consider if the model is appropriate to use for the given patient. Bayesian methods can be surprisingly robust and do not necessarily require an exact match between key characteristics in the model-building population and the patient, such as age and sex. However, the more deviation there is between population and patient, the less likely the model will be appropriate. Once a decision has been made that the model a priori seems appropriate, the next response is to carefully examine the inputs to the model. Potential sources of error contributing to bias include mistakes in the dose amounts documented and inaccurate recording of specimen collection times. If the inputs are deemed reliable, the next step is to consider whether the patient status has changed or is changing substantially to alter the accuracy of model predictions. For example, have interacting drugs been introduced which either lower or raise the monitored drug's concentrations? Is the patient's physiology changing? In cases where an alternative explanation for discordance is not evident, additional non-model-based strategies to individualize therapy should be considered.

5.3.2 | Rationale

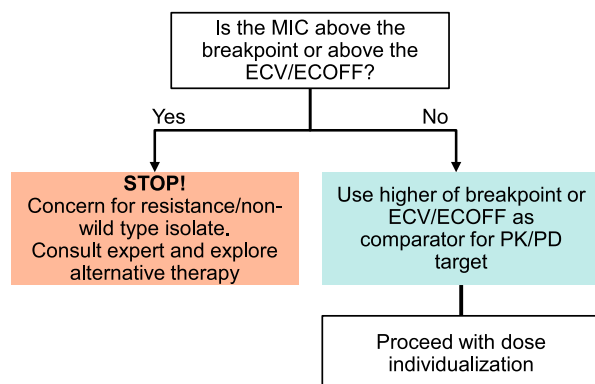
PK/PD targets for beta-lactam dose individualization take into consideration the pathogen MIC. Key terms and concepts related to antibacterial susceptibility testing are included in Table 2 and a flow diagram for selection of a PK/PD target based on susceptibility results is shown in Figure 2.

There is inherent variability in the MIC measurement [138] due to the method of testing, technical factors, the biology of the organism, and the extent of expression of phenotype [139]. In isolates without phenotypically detectable resistance to an antibiotic ('wild type'), MIC values can span 3–5 doubling dilutions. The ECV/ECOFF is a statistically defined threshold, representing the upper percentile of MIC values for wild-type isolates. It is used to distinguish wild-type from non-wild type populations. Isolates with MICs above the ECV/ECOFF are likely to have acquired resistance. Clinical breakpoints, established by experts, identify MIC cut-off values for which an isolate would be considered 'susceptible' or 'resistant' to a given antibiotic using typical doses [140, 141]. In practice, while a single MIC value is reported (e.g., 1 µg/mL) and compared to a breakpoint to guide therapy, this is an oversimplification. The average standard error of the reported MIC value is ± 1 doubling dilution (i.e., an MIC value

of 1 µg/mL may be 0.5 µg/mL or 2 µg/mL) and 95% of values are within ± 2 doubling dilutions (i.e., 0.25–4 µg/mL in this example) [142]. This variability can have a significant impact on the dose necessary for effectiveness. If the actual MIC is one doubling dilution higher than the measured (i.e., reported) MIC, twice as much beta-lactam antibiotic would theoretically be necessary to achieve comparable drug exposure.

There are no known clinical trials that compare different approaches to microbiologic target selection to guide individualized drug dosing [143]. For beta-lactam dose individualization, therefore, the panel supports use of the susceptibility breakpoint or the ECV/ECOFF (whichever is a higher numerical value) to determine the PK/PD target. Importantly, the MIC that appears in the patient's laboratory results may be below the susceptibility breakpoint/ECV/ECOFF, but the proposed approach accommodates the inherent MIC variability and avoids underprediction of the true MIC. The panel recognizes that this approach differs from what is currently undertaken at many centers and may require institutions to re-evaluate and modify procedures or re-train staff. In situations for which microbiologic data are not available (e.g., empiricism, culture negative infections) the breakpoint or ECV/ECOFF of the most likely pathogen(s) (e.g., *Pseudomonas aeruginosa* in nosocomial pneumonia) should

- 1 Determine the patient specific organism MIC, the susceptible breakpoint, and the ECV/ECOFF for the drug-organism pair
 - CLSI: <https://em100.edaptivedocs.net/Login.aspx> (for breakpoint)
 - EUCAST: https://www.eucast.org/mic_and_zone_distributions_and_ecoffs (for ECV/ECOFF)
- 2 Compare patient MIC to susceptible breakpoint and ECV/ECOFF



- 3 Use the algorithm to select the minimum suggested PK/PD target for the patient (minimum 100% $fT_{>MIC}$)

FIGURE 2 | Beta-lactam PK/PD target selection algorithm. Stepwise algorithm for determining the minimum pharmacokinetic/pharmacodynamic (PK/PD) target for beta-lactam therapy based on patient-specific minimal inhibitory concentration (MIC), susceptible breakpoint, and epidemiological cutoff value (ECV; I.e. ECOFF). The algorithm applies to scenarios where a patient-specific organism MIC is available. In cases for which microbiologic data are not available (e.g., empiricism, culture negative infections) the breakpoint or ECV/ECOFF of the most likely pathogen(s) should be used.

be used. Among studies evaluating the clinical impact of dose individualization (see Questions 1–4), the susceptibility breakpoint or the ECV/ECOFF was most commonly used for dosing decisions. Nevertheless, availability of an individual laboratory defined pathogen MIC may be preferred as such a measurement provides direct information regarding the degree of resistance.

Take for example the case of a patient with a *P.aeruginosa* bloodstream infection with an MIC of 0.25 µg/mL to meropenem reported in the patient's laboratory results. Using the reported MIC, this would mean a minimum target free beta-lactam concentration of 0.25 µg/mL for 100% of the dosing interval. The actual MIC may not measure exactly 0.25 µg/mL and there is a 95% probability that the concentration is somewhere between 0.06 and 1 µg/mL. The *P.aeruginosa* ECV/ECOFF for meropenem is 2 µg/mL and the susceptibility breakpoint is ≤ 2 µg/mL [140, 141]. Thus an MIC target of 2 µg/mL would be a reasonable reference value for $fT_{>MIC}$. Key patient-specific considerations would also need to be incorporated into clinical decision making (e.g., source of the bacteremia, risk for toxicity). In summary, in this circumstance, despite documented laboratory reporting of an MIC of 0.25 µg/mL, the clinical team would target a beta-lactam concentration of 2 µg/mL for 100% of the dosing interval to optimize treatment outcomes.

In situations where the reported MIC exceeds the ECV/ECOFF or the susceptibility breakpoint for a given antibiotic-organism combination, there is concern of antibiotic resistance, and the panel suggests exploring alternative antibiotics or antibiotic combinations with proven safety and effectiveness for the pathogen and infection type. Management of infections due to

drug-resistant organisms and the use of combination therapy is beyond the scope of the current guidance, but is addressed in detail elsewhere [51]. Expert consultation should be sought in cases where no antibiotic alternatives exist to evaluate the risks and benefits of selecting more aggressive PK/PD targets to effectively treat infections due to organisms with drug resistance. In the specific scenario with limited treatment options (i.e., resistance to first-line beta-lactams), the panel agreed that some experts would consider using one doubling dilution above the reported intermediate MIC as the PK/PD target, while others suggest setting the target based on the reported intermediate MIC. Competing factors such as the patient's baseline risk of toxicity, severity of illness, and site of infection must be considered in final decision making.

Genotype/phenotype discrepancies are beyond the scope of this document; however, these discrepancies should be discussed with the microbiology laboratory, infectious diseases experts, and repeat and/or further testing with alternate methods may be indicated [140].

5.4 | Question 13: What Are the Relevant Considerations for Incorporating Free (Unbound) Concentrations Into Beta-Lactam Individualization?

5.4.1 | Best Practice Statement

Where free beta-lactam concentrations are available and validated, these are preferred. The panel notes free beta-lactam

concentration assays are more technically complex, expensive, and not widely available. When measurement of free beta-lactam concentrations is unavailable, the panel supports estimating free concentrations from total concentrations using published protein binding values from the literature.

5.4.2 | Rationale

Only free (unbound) beta-lactam drug can cross membranes and exert a pharmacologic effect. Although free beta-lactam concentrations are more often associated with therapeutic (i.e., $fT > MIC$) and toxic effects than total concentrations, total concentrations are more widely measured in clinical practice. Measurement of free drug concentrations requires additional time and resources, and is a more technically complex process, as methods required for such measurement (i.e., ultrafiltration) are susceptible to experimental variation due to pH, temperature, or membrane type [144, 145]. Highly protein bound beta-lactams (e.g., ceftriaxone, ertapenem, cefazolin, flucloxacillin) may be best suited for this additional resource investment [146, 147]. Estimated free beta-lactam concentration is suitable for PK/PD evaluation in drugs with limited protein binding (e.g., meropenem, cefepime) or when only total concentration assays are available (Table 2). In certain circumstances the relationship between total and free concentrations may be unreliable [148]. Protein binding may vary in patients with critical illness, hypoalbuminemia, who are pregnant, or neonates [149, 150]. In these populations, reduced protein binding leads to a higher proportion of free (unbound) drug [151, 152]. Early in the dosing interval, transient increases in absolute free concentrations from low protein binding could potentially improve bacterial killing. However, depending on the extraction ratio, an increased free drug fraction may lead to enhanced clearance, potentially resulting in subtherapeutic exposures. While PK/PD targets are ideally defined in terms of the unbound (free) beta-lactam concentration, especially for highly protein-bound drugs, the published literature often does not clearly distinguish between total and unbound concentrations when describing PK/PD target optimization. The inconsistency, in part, stems from efforts to generalize a singular PK/PD target across multiple beta-lactams, despite variability in protein binding between drugs and among patients.

5.5 | Future Directions on How to Achieve Beta-Lactam Individualization

Existing up-front dosing with MIPD largely relies on population PK models. Future studies could explore the use of artificial intelligence to streamline PK model development, or as an adjunct (i.e., hybrid PK-machine learning models) or alternative to population PK to guide beta-lactam dose individualization. To date this has only been sparingly evaluated for beta-lactams [153–155]. We envision studies which test the role for artificial intelligence to predict patient response to therapy, suggest optimal drug dosing, and streamline the workflow for beta-lactam dose individualization. Acceptability of artificial intelligence-guided beta-lactam dosing models will need to be evaluated, given prior documented skepticism among clinicians with similar tools for other antibiotics [156].

Heterogenous approaches to MIC determination [138, 157] in beta-lactam dose individualization studies including the use of measured MICs, susceptibility breakpoints, and ECV/ECOFF, makes aggregation of data across studies difficult. In addition to clarity in reporting, future research will also need to test different methods for MIC determination for beta-lactam dose individualization. Preliminary evidence suggests that use of different MIC parameters could substantially impact reported PK/PD target attainment [143]. Future research should also explore validation and application of beta-lactam concentration analysis in matrices other than plasma/serum including at the site of infection, the clinical relevance and practical utility of assays that measure free (unbound) beta-lactam concentrations, and the use of minimally or non-invasive sampling approaches (e.g., dried blood spots or volumetric absorptive microsamples) [39, 158, 159].

6 | Special Considerations

6.1 | Resource-Limited Settings

It is important to note that most of these data are obtained from resource rich environments. Future studies should be performed in resource limited settings where patient population, drug selection, dosing norms, susceptibility patterns, practice model structure, and access to software and beta-lactam concentration assays differ. The panel acknowledges the substantial complexity of implementing a beta-lactam dose individualization program in the absence of adequate human and technical resources.

6.2 | Children/Neonates

Pediatric patients treated with beta-lactams range in age from pre-term neonates to young adults. Across this age spectrum there are broad differences in PK that affect beta-lactam behavior [160–162]. Younger patients, particularly neonates and infants, have a greater proportion of total body water and immature end organs, which could lead to a larger volume of beta-lactam distribution and lower clearance, relative to adults, relatively increasing the time above MIC during the dosing interval. Once organs fully mature, clearance in pediatric patients may be more rapid than adults, when normalized to body size, which could in turn increase the risk for under exposure. Pediatric patients also have a relatively higher prevalence of congenital disease states such as cystic fibrosis which alter PK and increase the risk for infections from drug-resistant pathogens, making these patients ideal candidates for beta-lactam dose individualization. Fewer beta-lactam PK models have been developed for pediatric patients and rarely are these available in MIPD software programs. For maximal generalizability, models should reflect pediatric patients across the age spectrum and include a maturation factor. Implementation of TDM in pediatric patients is also distinct from adults. New extracorporeal organ support devices used in this population have limited data on the impact on beta-lactam PK [163]. Pediatric patients often have blood volume constraints and limited intravenous access which can add complexity to specimen collection and dosing regimen determination. Different outcomes in trials may need to be considered for pediatric patients,

since mortality is lower than for adults. Outcomes including antibiotic escalation, need for more diagnostic evaluations (e.g., number and volume of blood samples), length of fever are all potentially highly relevant for this subpopulation [164].

7 | Conclusion

The beta-lactam dose individualization consensus guidance provides evidence-based, yet practical recommendations to support individualized patient care. Beta-lactam dose individualization, where measured concentrations are used to individualize therapy for a given patient, is a promising strategy to improve outcomes in selected populations. Current evidence suggests potential effectiveness benefits of beta-lactam dose individualization, especially for clinical cure; however, there are few high-quality clinical trials. Beta-lactam dose individualization has not been linked with worsening adverse effects and promotes more rational antibiotic use aligned with antimicrobial stewardship goals. Implementation should be tailored to local resources, including access to drug assays, MIPD tools, and multidisciplinary input and expertise. Future research should focus on identifying optimal PK/PD targets in well-defined patient populations, evaluating patient-centered effectiveness and safety outcomes including collateral impact on the host microbiome, investigating the heterogeneity of treatment effects, identifying key subgroups uniquely likely to benefit from beta-lactam dose individualization, and assessing cost-effectiveness.

Author Contributions

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Conflicts of Interest

Personal and financial relationships with industry and other entities are disclosed for all members of the panel in Appendix S1.

Data Availability Statement

The data that supports the findings of this study are available in Appendix S1 of this article.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix S1:** Conflict of Interest Disclosures. **Appendix S2:** Literature search and evidence review. **Appendix S3:** Meta-analysis methods. **Appendix S4:** Evidence synthesis including meta-analyses. **Table S1:** Q1—What is the impact of beta-lactam dose individualization on MORTALITY? **Table S2:** Q1—What is the impact of beta-lactam dose individualization on CLINICAL CURE? **Table S3:** Q1—What is the impact of beta-lactam dose individualization on MICROBIOLOGIC CURE? **Table S4:** Q1—What is the impact of beta-lactam dose individualization on HOSPITAL LENGTH OF STAY? **Table S5:** Q1—What is the impact of beta-lactam dose individualization on ICU LENGTH OF STAY? **Table S6:** Q2—What PK/PD targets are preferred for beta-lactam individualization to improve treatment outcomes? **Table S7:** Q3—What is the impact of beta-lactam dose individualization on patient-centered SAFETY OUTCOMES? **Table S8:** Q4—Summary of safety endpoints based on beta-lactam exposure thresholds across included studies. **Appendix S5:** Main components of integrating beta-lactam dose individualization into the clinical.